Retail Sales Data Analytics & Business Intelligence

InternNova Week 6 Capstone | Atanu Manna

1. Project overview

This project analyzes transaction-level retail data to understand revenue by product, country and time. Python and Pandas perform cleaning and EDA; Matplotlib generates six charts.

2. Dataset and attribution

Source: UCI Machine Learning Repository, Online Retail. Original dataset: 541,909 rows and 8 columns. Dataset period: 1 December 2010 to 9 December 2011. Citation: Chen, D. (2015). *Online Retail* [Dataset]. UCI Machine Learning Repository. DOI: 10.24432/C5BW33. License: CC BY 4.0.

3. Data preparation

Removed 5,268 exact duplicate rows. Flagged 9,251 cancellation invoices. The positive-sales analysis contains 524,878 rows after excluding cancellations, non-positive quantities/prices, invalid dates and missing descriptions. Missing CustomerID values are not filled with invented identities. Revenue = Quantity × UnitPrice in GBP.

4. EDA summary

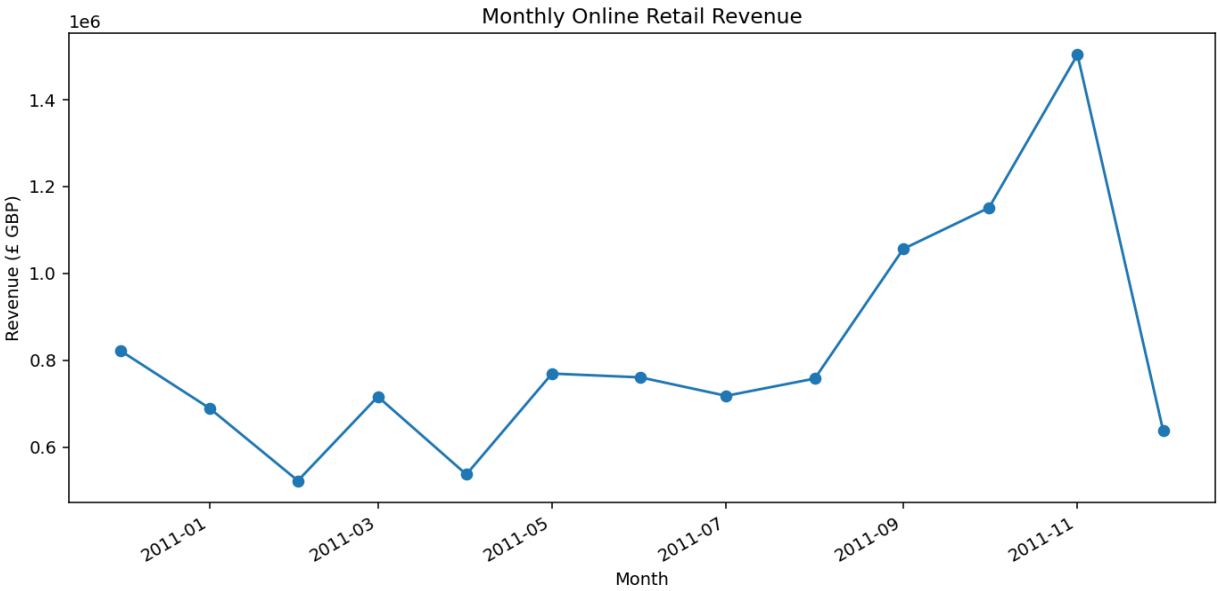
Metric	Verified result
Positive-sales revenue	£10,642,110.80
Unique invoices	19,960
Customers with ID	4,338
Unique stock codes	3,922
Line revenue outlier rows (1.5×IQR)	42,624
Highest revenue country	United Kingdom: £9,001,744.09
Top product	DOTCOM POSTAGE: £206,248.77

Outliers are flagged, not automatically removed. Revenue is not profit because the dataset contains no cost field.

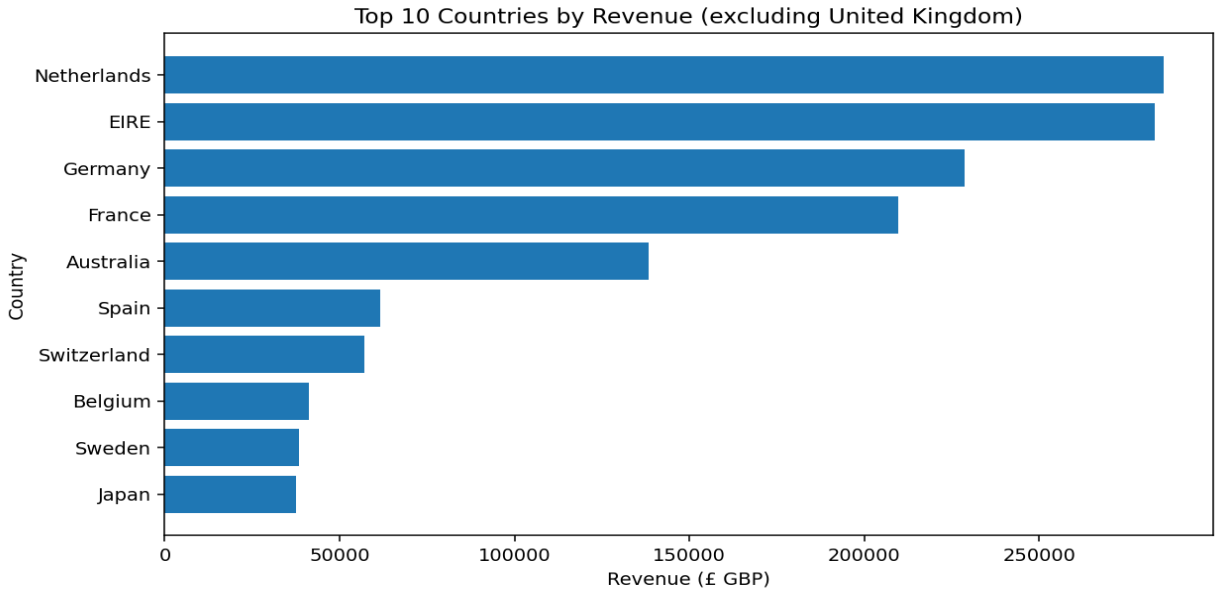
5. Visualizations

Six charts generated from the cleaned positive-sales data: monthly revenue, top countries excluding UK, top products, quantity versus line revenue, correlation heatmap, and weekday revenue.

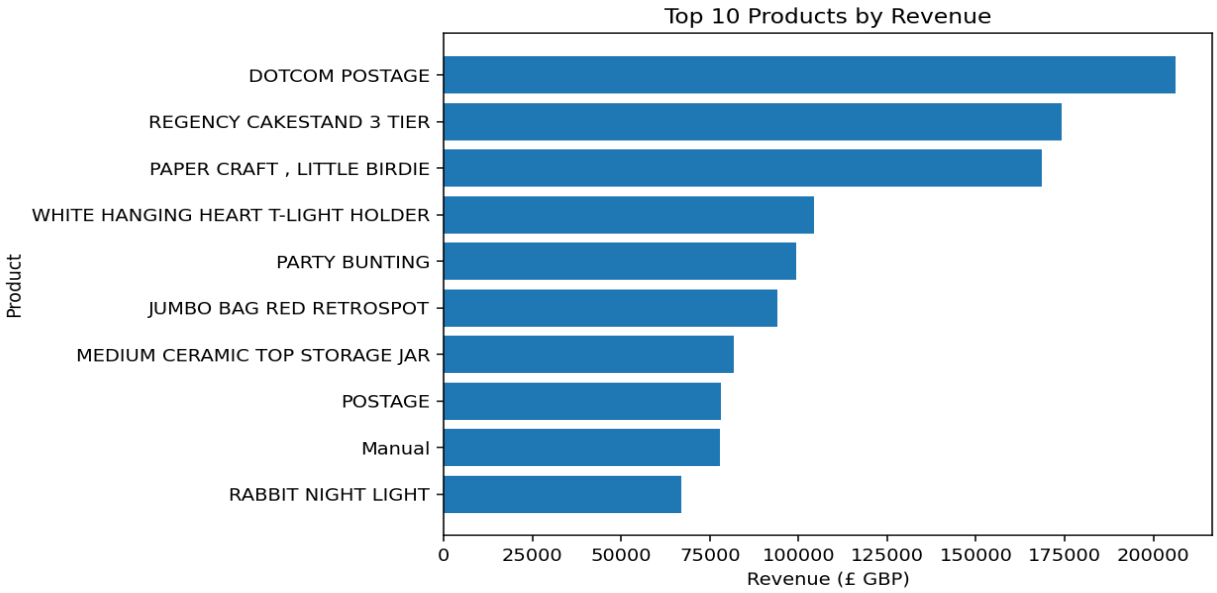
Monthly revenue



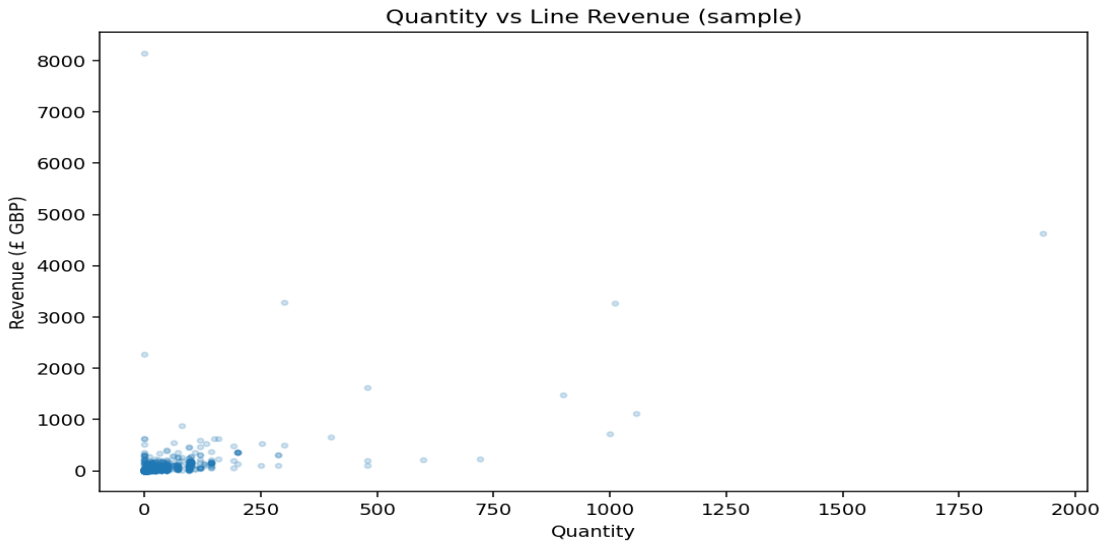
Top countries excluding UK



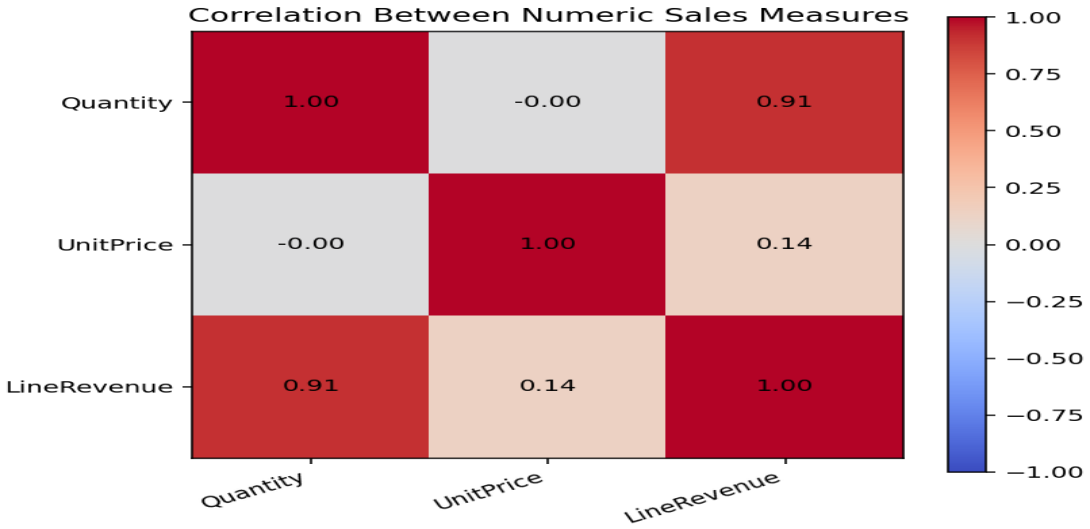
Top products


Top 10 Products by Revenue
DOTCOM POSTAGE
REGENCY CAKESTAND 3 TIER
PAPER CRAFT , LITTLE BIRDIE
WHITE HANGING HEART T-LIGHT HOLDER
PARTY BUNTING
JUMBO BAG RED RETROSPOT
MEDIUM CERAMIC TOP STORAGE JAR
POSTAGE
Manual
RABBIT NIGHT LIGHT
Product
0
25000
50000
75000
100000
125000
150000
175000
200000
Revenue (£ GBP)

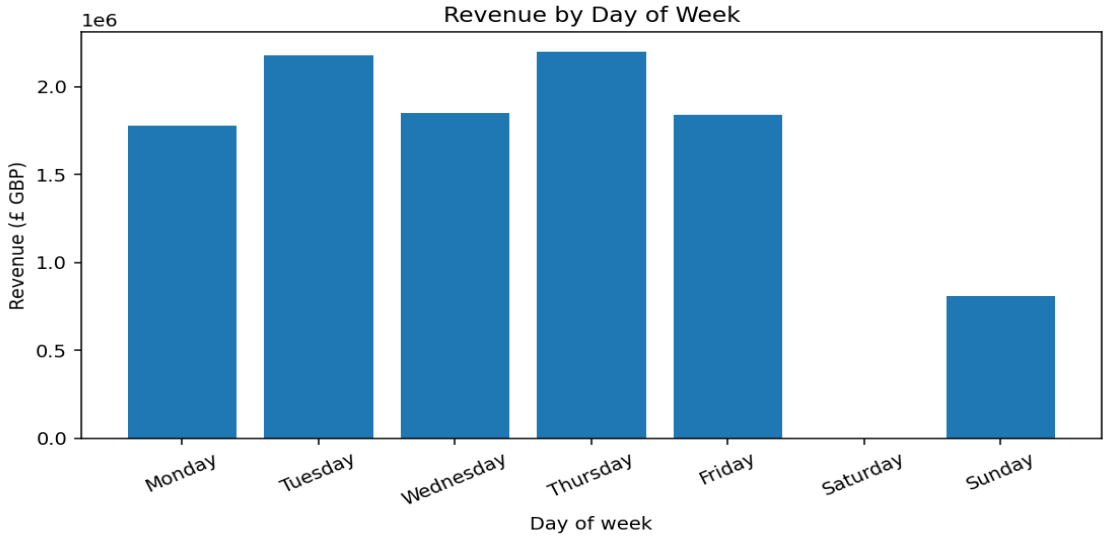
Quantity versus line revenue



Correlation heatmap



Revenue by weekday



6. Power BI dashboard

The native Power BI dashboard must be created in Power BI Desktop. Import data/processed/cleaned_sales.csv. Add KPI cards for Total Revenue, Total Invoices and Unique Customers; visuals for monthly revenue, top products, country revenue and weekday revenue; slicers for Country, InvoiceDate and product. Save as powerbi/Retail_Sales_Dashboard.pbix and insert genuine screenshots.

7. Evidence-based insights

1. Highest revenue country: United Kingdom (£9,001,744.09).

2. Top product by line revenue: DOTCOM POSTAGE (£206,248.77).

3. Highest revenue month: 2011-11 (£1,503,866.78).

4. Lowest revenue month: 2011-02 (£522,545.56); partial months can affect comparisons.

5. Highest revenue weekday: Thursday (£2,199,292.57); lowest: Sunday (£806,790.78).

6. The correlation heatmap summarizes numeric associations; correlation does not prove causation.

8. Recommendations

1. Review stock availability and merchandising for DOTCOM POSTAGE; validate profitability with cost data, which is not present here.

2. Investigate the leading market (United Kingdom) and compare shipping cost, customer retention and margin before reallocating budget.

3. Use monthly and weekday patterns for inventory planning while accounting for seasonality and incomplete months.

9. Limitations and completion status

The dataset is historical (2010–2011), not current market performance. This report contains Python-derived results and charts. The real .pbix dashboard, genuine Power BI screenshots, GitHub repository URL and completed commit history still need to be added before final submission.